I. Overall Requirements

1. Introduction

The purpose of this project is to build a learning support system named StudyHub from that teachers or group managers can collect, consolidate, and share their knowledge to others (in Member user class) for their studying/revising anywhere or anytime. With that purpose, the StudyHub application would have following functionalities:

1. For Member (learners or potential learners):

- Register user account and login with that account.
- Browse available courses (groups of shared knowleges).
- Select and provide reason enroll the course in his/her interest.
- View list of courses he/she is allowed to join (approved or registered directly by Manager)
- View the course studying progress, browse and study/re-study the course lessons
- View and update his/her profile (including changing the login password)

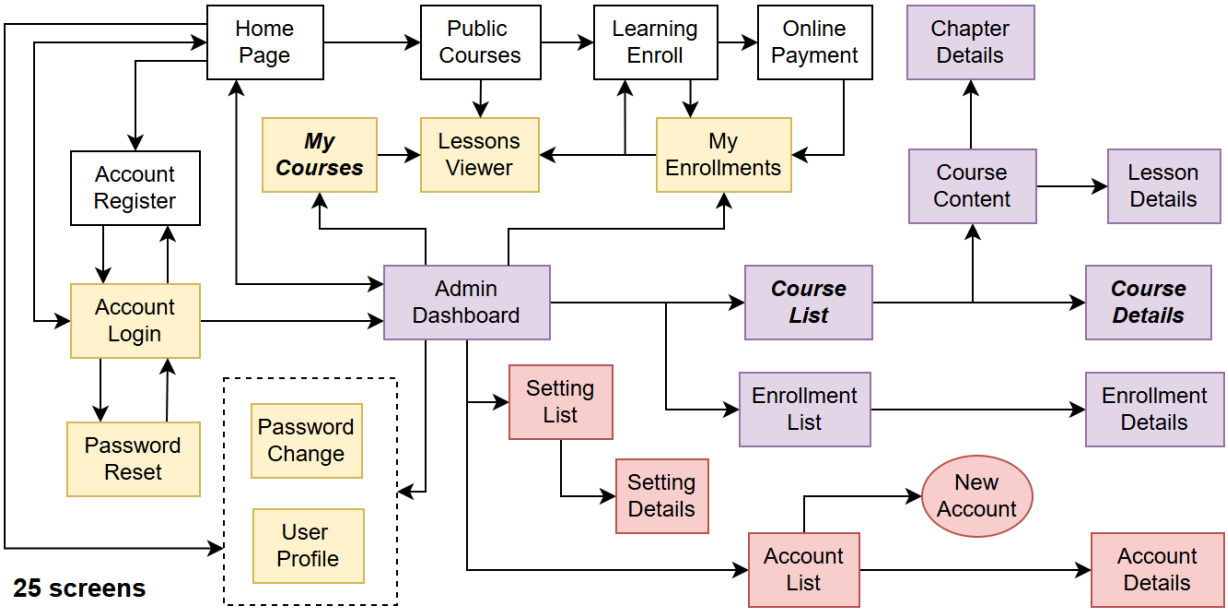
2. For Manager (teachers or course managers who is responsible for the assigned courses):

- Have all functionalities of the Member as listed above.
- Browse the list of in-charged courses (the courses that he/she have created or been assigned).
- Manage course details: register new course, view, delete or update course details.
- Manage course contents: add new, view, update or delete course chapters, chapter lessons
- Manage course enrollments: view enrollment list, add/import/export enrollment(s), view or update enrollment details (approve or reject relevant rejection notes).

3. For Administrator (the persons who manages and operates the application):

- Have all functionalities of the Manager as listed above.
- View system dashboard and maintain system settings: system configs, course categories,..
- Manage user accounts: view account list, add new account, update or delete account details.
- Manage system settings: view setting list, add new setting, update or delete setting details.

2. Screen Flow



3. Screen Descriptions

Screen	Description
Home Page	Show featured (hot) active courses
Public Courses	Show all active courses, sorted by course name
User Register	Allow new user to register Member account to access the system; After registering, user need to verify his/her email by clicking the link in the account verifying email from the system. After clicking the verifying link, user is redirected to the User Login page with appropriate toast message
User Login	Authenticate user and give him/her accessing system's functionalities; Inactive and unverified user is not allowed to login. For unverified user, he/she would have option to request new account verifying email to verify his/her account
Password Reset	Allow user to reset his/her password for the case he/she forgot it; After user inputs valid email address, system would send user an email with the password-resetting link from that he/she can access the page for inputting new password; The password-resetting link is expired after 30 minutes.
User Profile	User can view and update his/her profile details
Password Change	Allow user to change his/her password; He/she would be logged out and redirected to the User Login page after successfully changing the password
Admin Dashboard	Admin and Manager can view full system statistics and latest courses. Admin can have the links to the Course Details and Course Content pages in all latest courses. However, with Manager role, those links are limitted to his/her assigned courses.
Setting List	View and activate/deactivate the system settings
Setting Details	Add new or update a specific system settings. The setting name must be unique within each group of system settings with the same setting type.
User List	View, delete or activate/deactivate the user accounts. Only new user (who has not had any transactions yet in the system) can be deleted
New User	Create new user, then generate a random password and send to newly-created user's email along with other account-creation notification
User Details	View and/or update details of a specific user account

Course List	View, delete or publish/unpublish the courses. Only new course (which are note related to any other system transactions in the system) can be deleted
Course Details	View, add new, or update course details. Manager can have accesses only to his/her assigned courses and he/she are not allowed to change the course price and status (published, unpublished)
Course Content	View details of the course and course contents (chapters, lessons); Manager has only accesses to the courses that he/she has been assigned; On this page, user can choose to delete a specific chapter or lesson or have links/buttons to access the Chapter Details and Lesson Details pages for adding new or updating details of specific chapter, lesson.
Chapter Details	Allow user to provide details for adding new or updating existing chapter; Manager has only accesses to the chapters of the courses that he/she has been assigned;
Lesson Details	Allow user to provide details for adding new or updating existing lesson; Manager has only accesses to the lessón of the courses that he/she has been assigned;
Learning Enroll	Both Guest and Member can enroll a specific course (if he/she has not been approved to join that course before). For the case of Member, his/her profile information would be automatically filled into the registration fields, and he/she can update to enroll the course for other people if neeeed. After successfull registered, a notification email would be sent to both registering and registered people informing about the success registration; If registered user does not exist, he/she would be registered into the system with newly-random access password and that password is sent to the registered user (along with the registering notification email)
Online Payment	This is for user to handle the bank transfering (using PayOS feature) or internet banking (using VnPay feature) payment according to the payment method that he/she has chosen from the previous step (Learning Enroll screen) and update to the enrollment status as approved for success payment
My Enrollments	Allow user to view his/her enrollments; For the approved enrollments, user can choose to view the enrollment details and registerd user's learning progress in a modal dialog. For the pending or rejected enrollments, user can choose to go back to the Learning Enroll screen to view rejection notes (if any), then updating and making payment.
Enrollment List	View, export the list of enrollments for the courses and import Excel enrollment list for a specific course; The user can also delete his/her created enrollments and approve or reject a specific enrollment (with relevant notes) The Manager can have accesses to enrollments of his/her assigned courses only while the Admin can have accesses to enrollments of all the courses
Enrollment Details	View, add new or update details for a specific enrollments; The Manager can have accesses to enrollments of his/her assigned courses only while the Admin can have accesses to enrollments of all the courses
My Courses	Allow user to view the courses that he/she has been approved to join
Lesson Viewer	User can view all the active chapters and lessons of the selected course; The chapters and lessons in each chapter are sorted in the way Admin or Manager has defined before. Each lesson is shown in the suitable form (video player, PDF viewer, or rich-text post) depending on its content type.

4. Screen Authorization

System provide the functionalities or accesses to 4 different user classes as listed in the table below. Details are provided in the description for each screen in the previous section.

Function	Admin	Manager	Member	Guest
Home Page	x	x	x	x
Public Courses	x	x	x	x
User Register	x	x	x	x
User Login	x	x	x	
Password Reset	x	x	x	
User Profile	x	x	x	
Password Change	x	x	x	
Admin Dashboard	x	x		
Setting List	x			
Setting Details	x			
User List	x			
New User	x			
User Details	x			
Course List	x	x		
Course Details	x	x		
Course Content	x	x		
Chapter Details	x	x		
Lesson Details	x	x		
Learning Enroll	x	x	x	x
Online Payment	x	x	x	x
My Enrollments	x	x	x	
Enrollment List	x	x		
Enrollment Details	x	x		
My Courses	x	x	x	
Lesson Viewer	x	x	x	

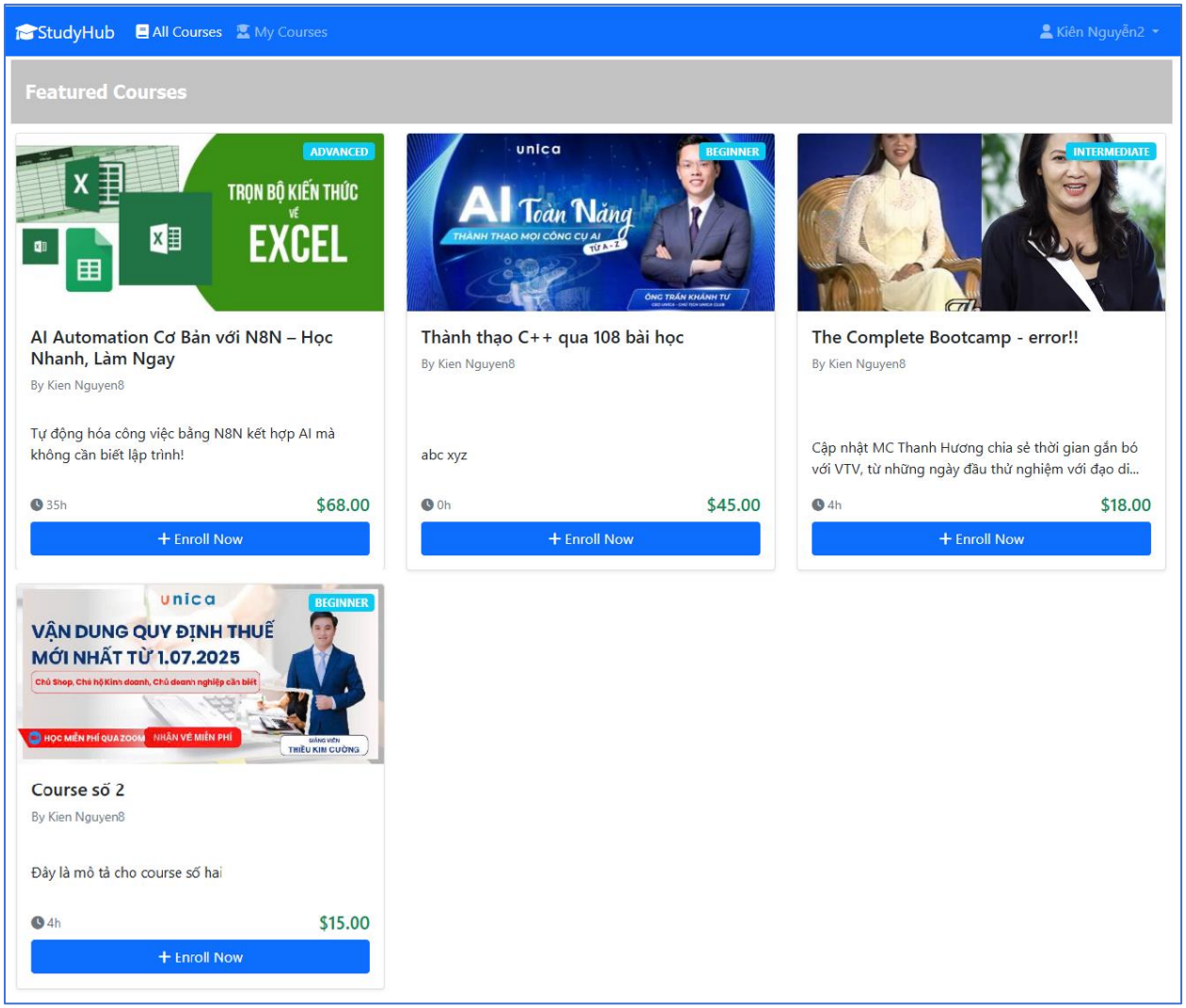
5. Main Workflows

Function	System Administration	User Authentication	Course Management	Course Enrollment	Lesson Access
Home Page	x	x	x	x	x
Public Courses	x		x	x	
User Register		x			
User Login	x	x	x		x
Password Reset		x			
User Profile	x	x			
Password Change		x			
Admin Dashboard			x		
Setting List	x				
Setting Details	x				
User List	x	x			
New User		x			
User Details	x	x			
Course List	x		x		x
Course Details	x		x		x
Course Content					x
Chapter Details					x
Lesson Details					x
Learning Enroll				x	
Online Payment				x	
My Enrollments			x	x	
Enrollment List				x	
Enrollment Details				x	
My Courses			x	x	x
Lesson Viewer					x

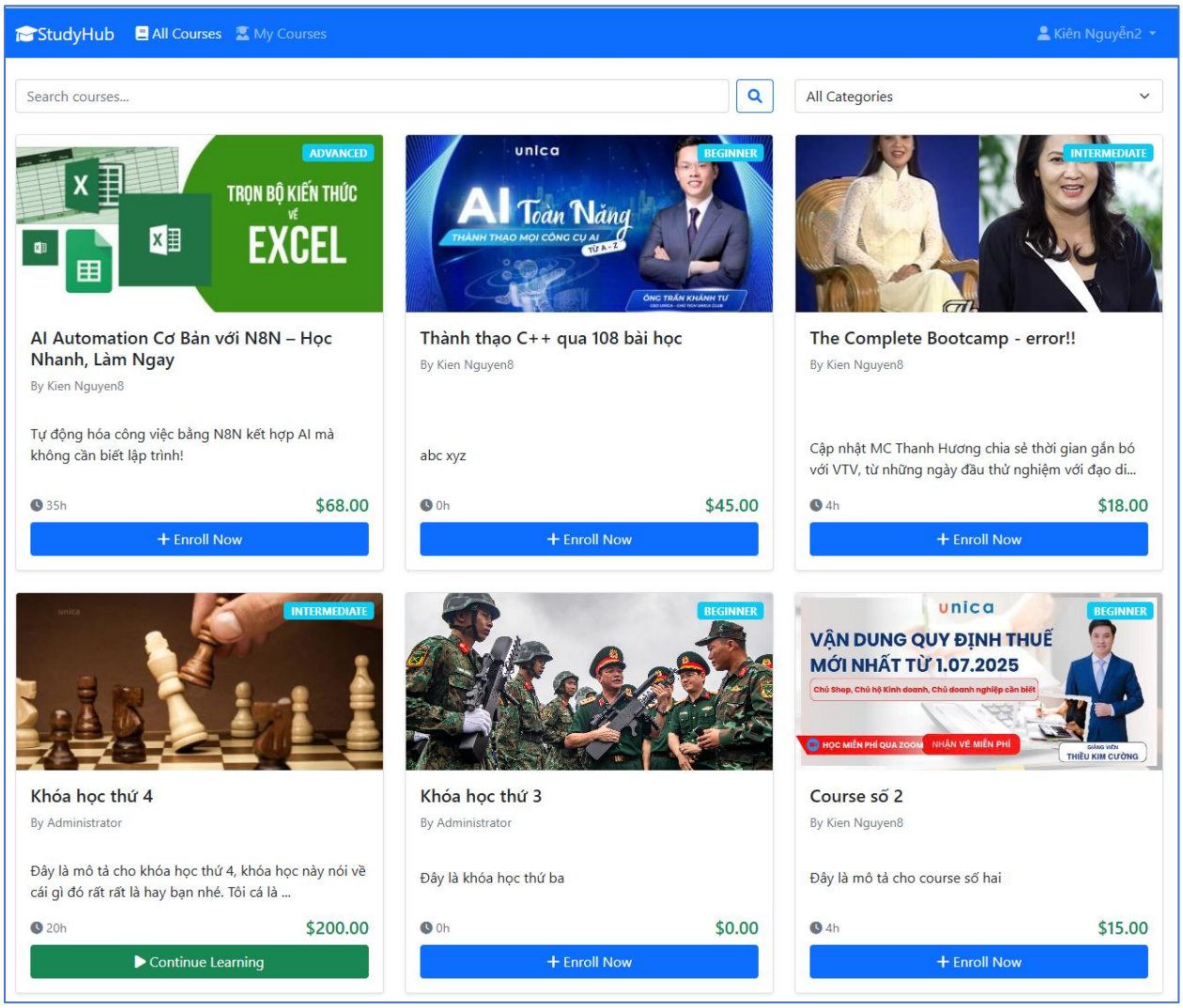
II. Screen Layouts

1. Common Feature

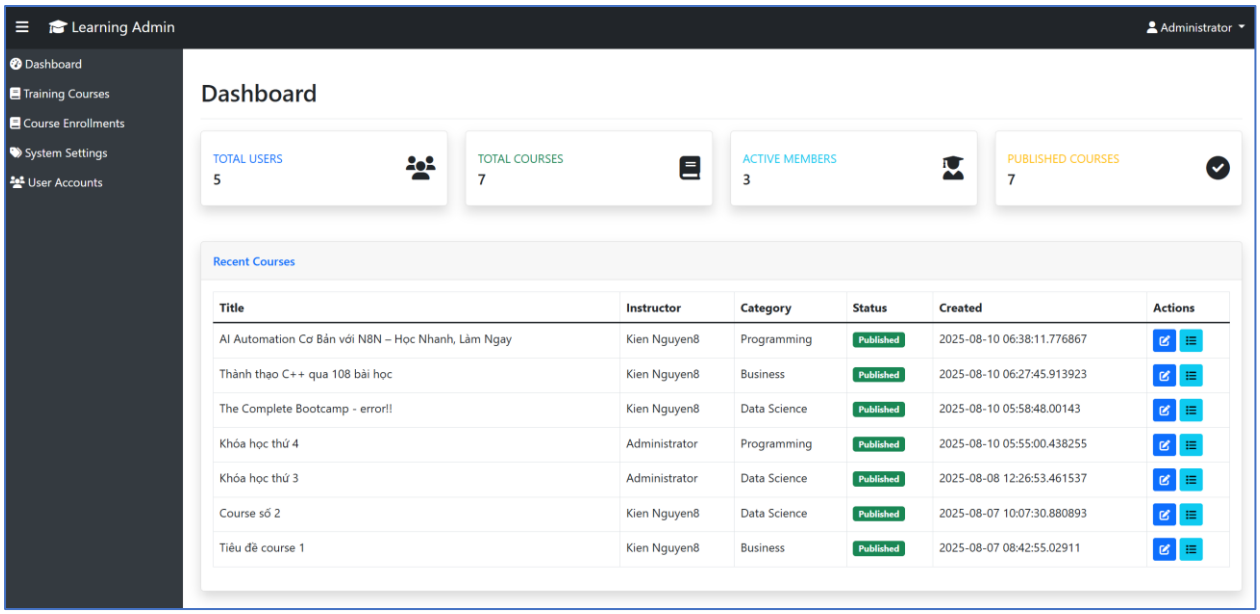
1.1. Home Page



1.2. Public Courses

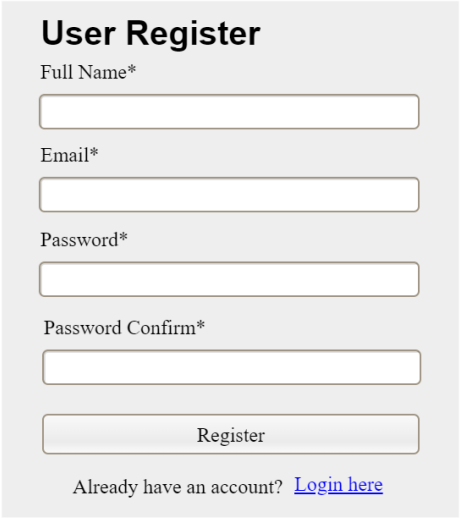


1.3. Admin Dashboard

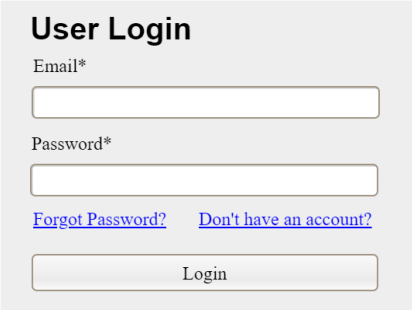


2. User Authentication

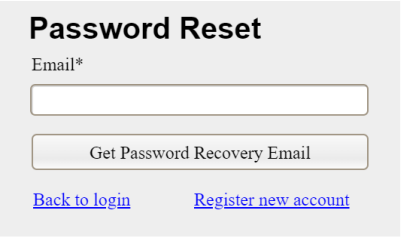
2.1. User Register

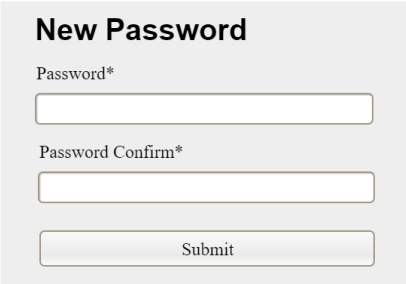


2.2. User Login

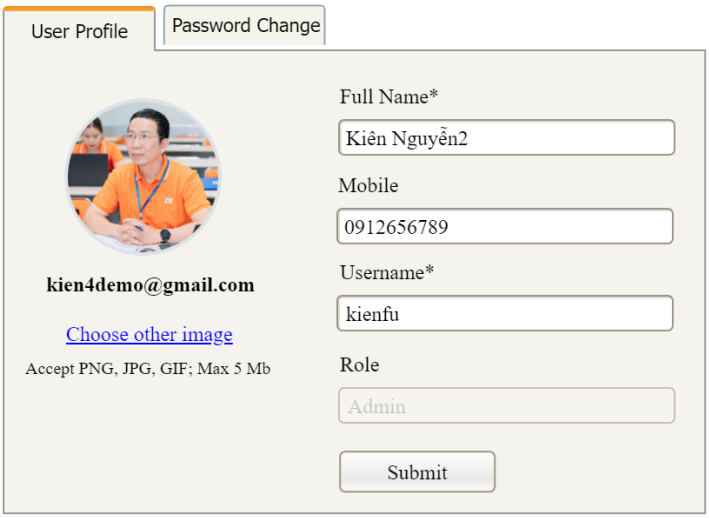


2.3. Password Reset

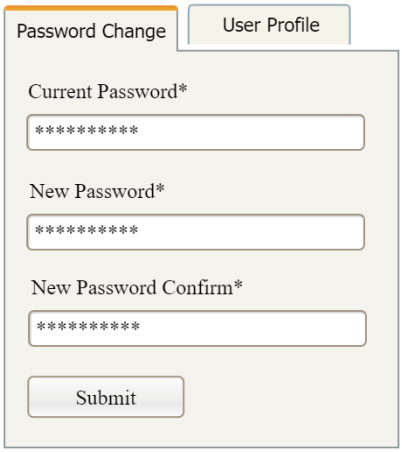




2.4. User Profile



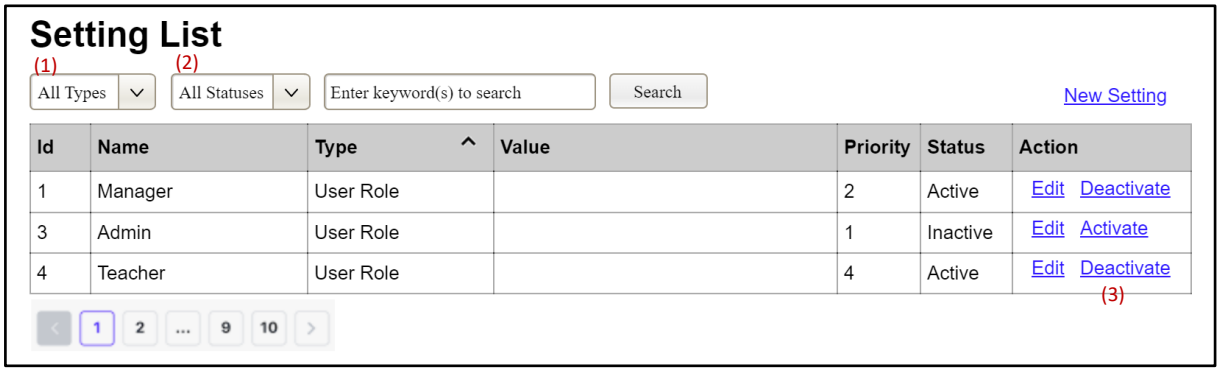
2.5. Password Change



3. System Administration

3.1. System Setting

a. Setting List



This screen allows the Administrator to:

- View Setting List: view list of current master data.
- Filter Setting List: filter master data by data types, statuses
- Search Settings: enter keyword(s) to search master data by their names or values
- Sort Setting List: sort master data list (ascending, descending) by clicking column headers

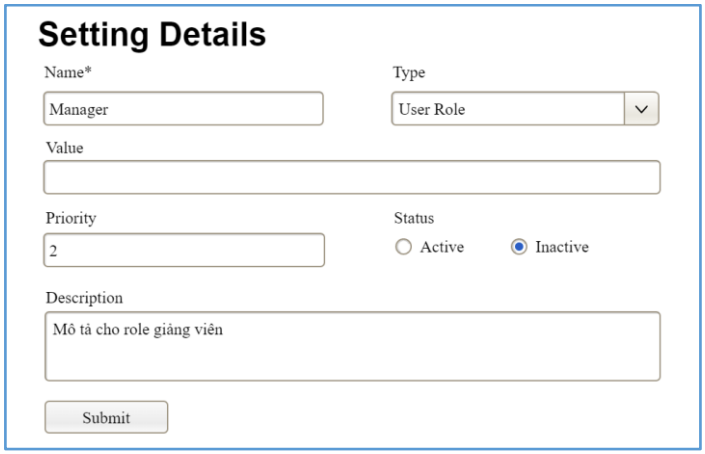
On the screen, s/he can also

- Activate/Deactivate Setting: change status of a specific inactive/active master data
- Choose to go to the Setting Details screens for adding new or updating an existing master data by clicking the New Setting or Edit link.

Field Description

Field Name	Description
(1)	Initial values: all the active setting names with null or blank type Hover the mouse to show the field name: “Setting Type”
(2)	Initial values: All Statuses, Active, Inactive (default value “All Status”) Hover the mouse to show the field name: “Setting Status”
(3)	The change-status action is Activate or Deactivate depending on the current status of the relevant setting (Inactive or Active, respectively).

b. Setting Details



This screen allows the Administrator to:

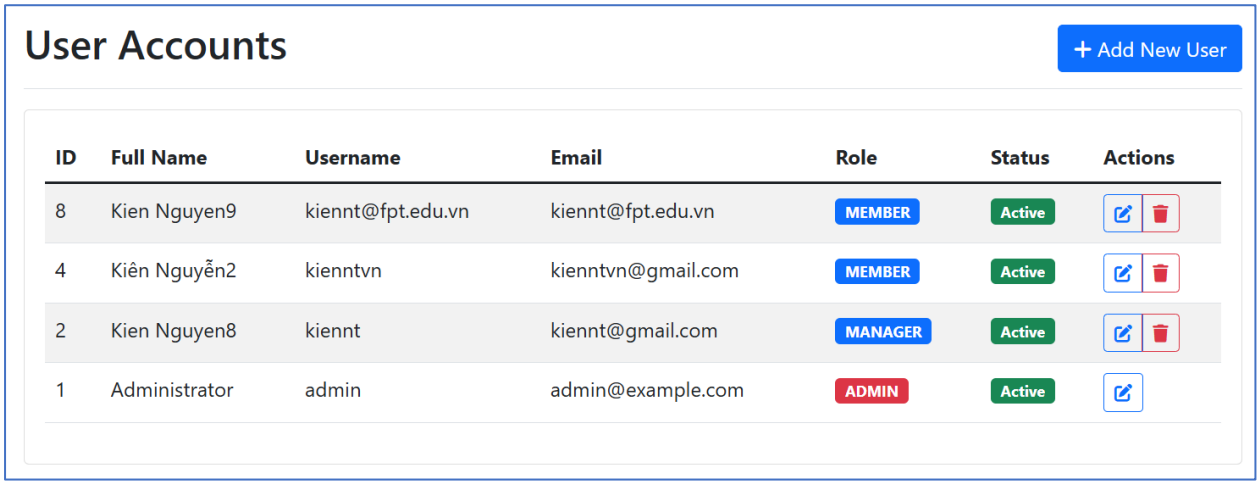
- Add New Setting: add new master data.
- Update Setting Details: update details of a specific master data

Field Description

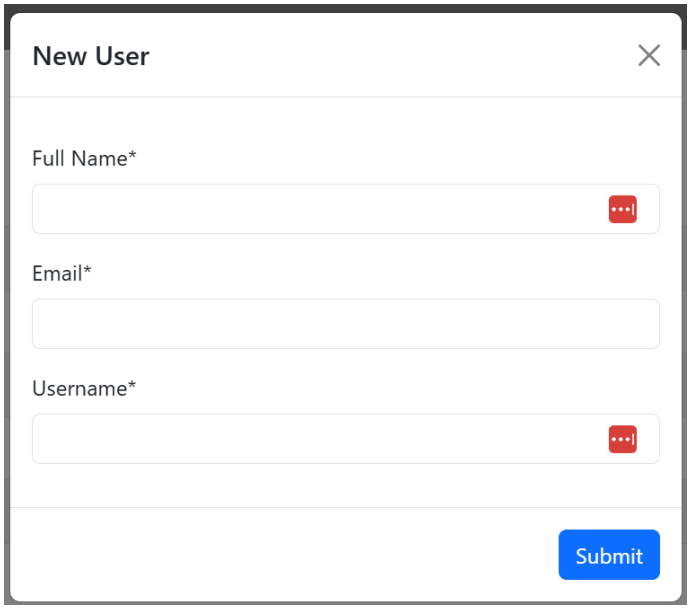
Field Name	Description
Name	Data type: non-digit string, max length of 20 characters Constraints: unique value in the selected type
Type	Initial data values: all active setting names (with null or blank type)
Value	Data type: any string, max length of 100 characters
Priority	Data type: a positive integer
Description	Data type: any string, max length of 200 characters

3.2 User Management

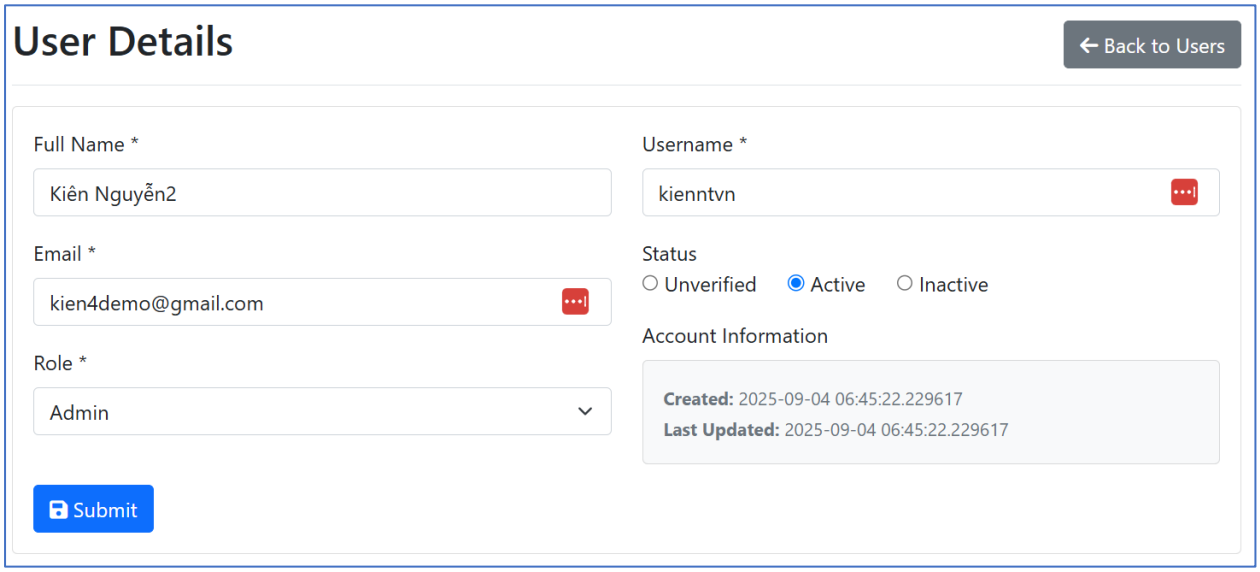
a. User List



b. New User

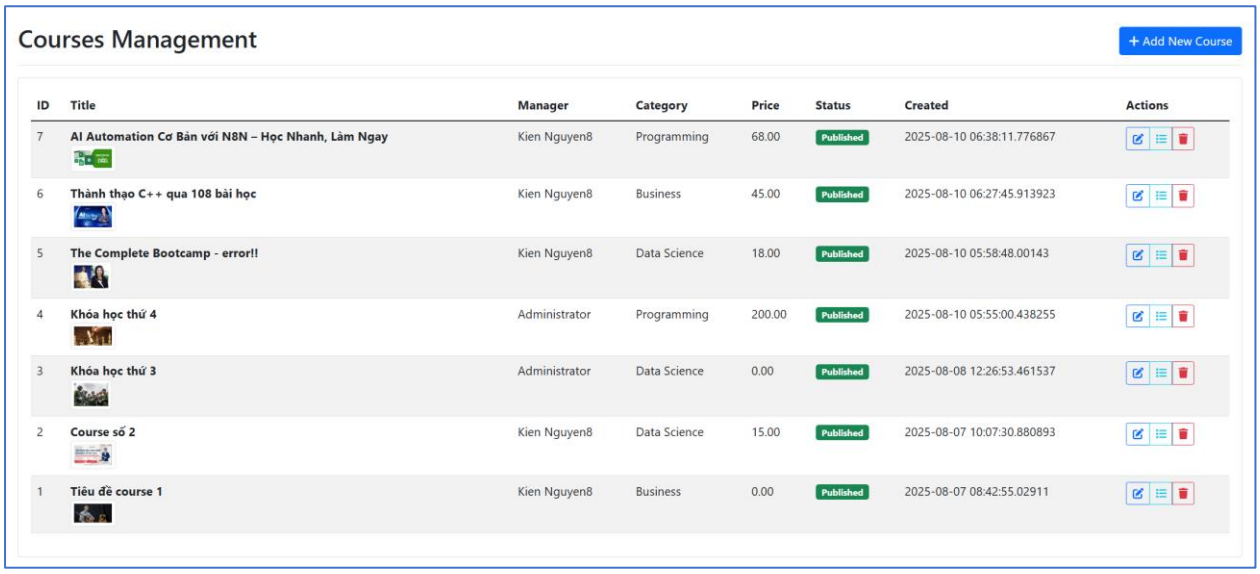


c. User Details



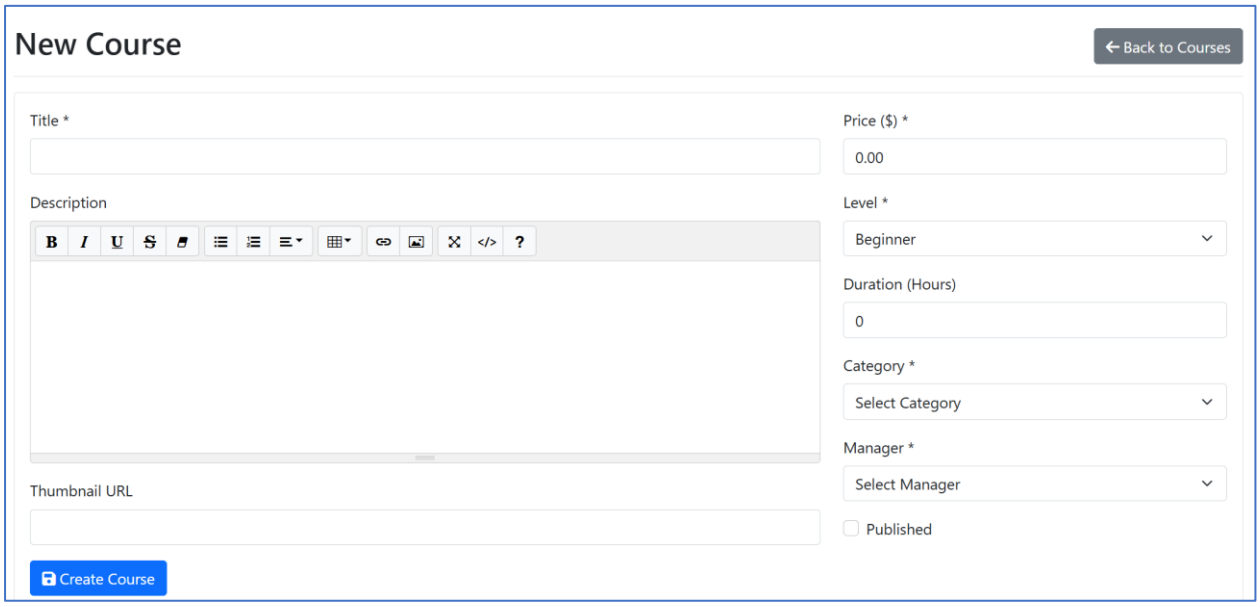
4. Course Management

4.1. Course List

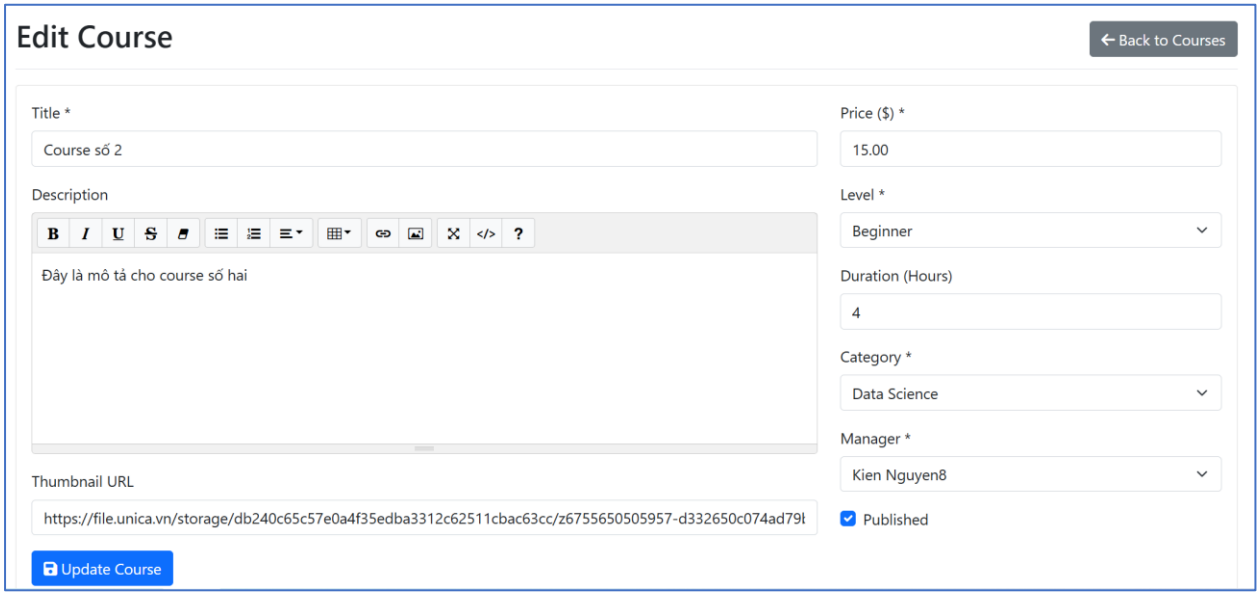


4.2. Course Details

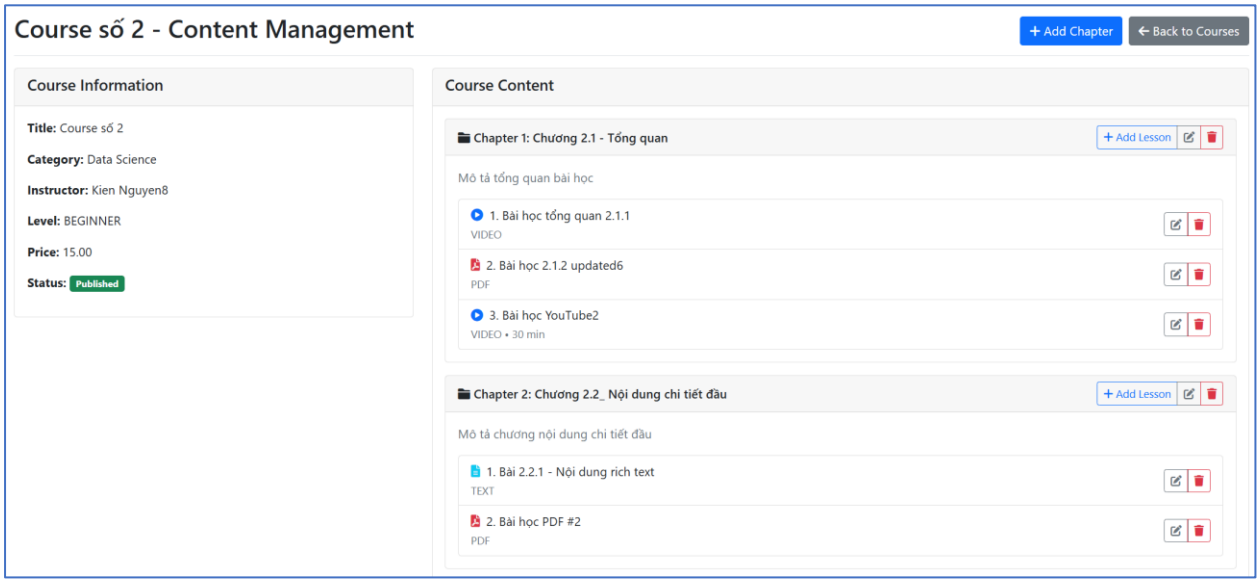
a. New Course



b. Course Edit

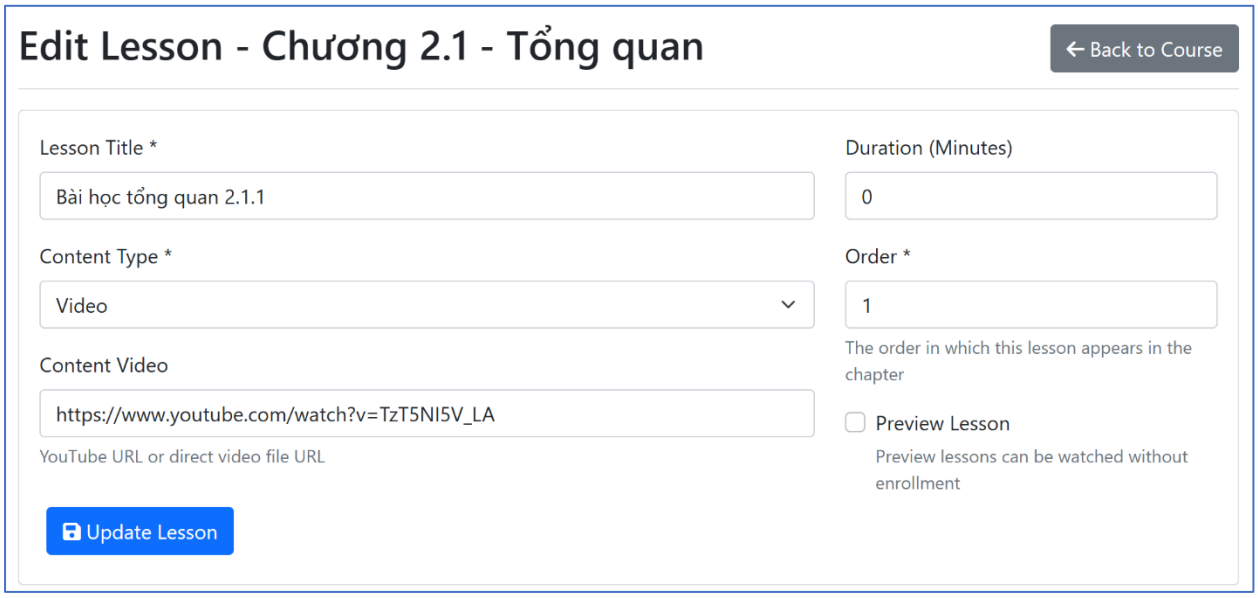


4.3. Course Content



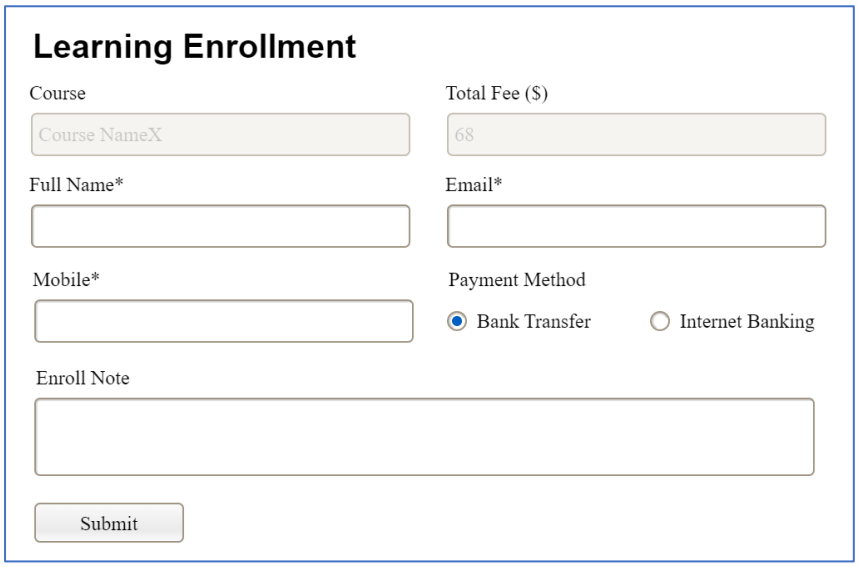
4.4. Chapter Details

4.5. Lesson Details



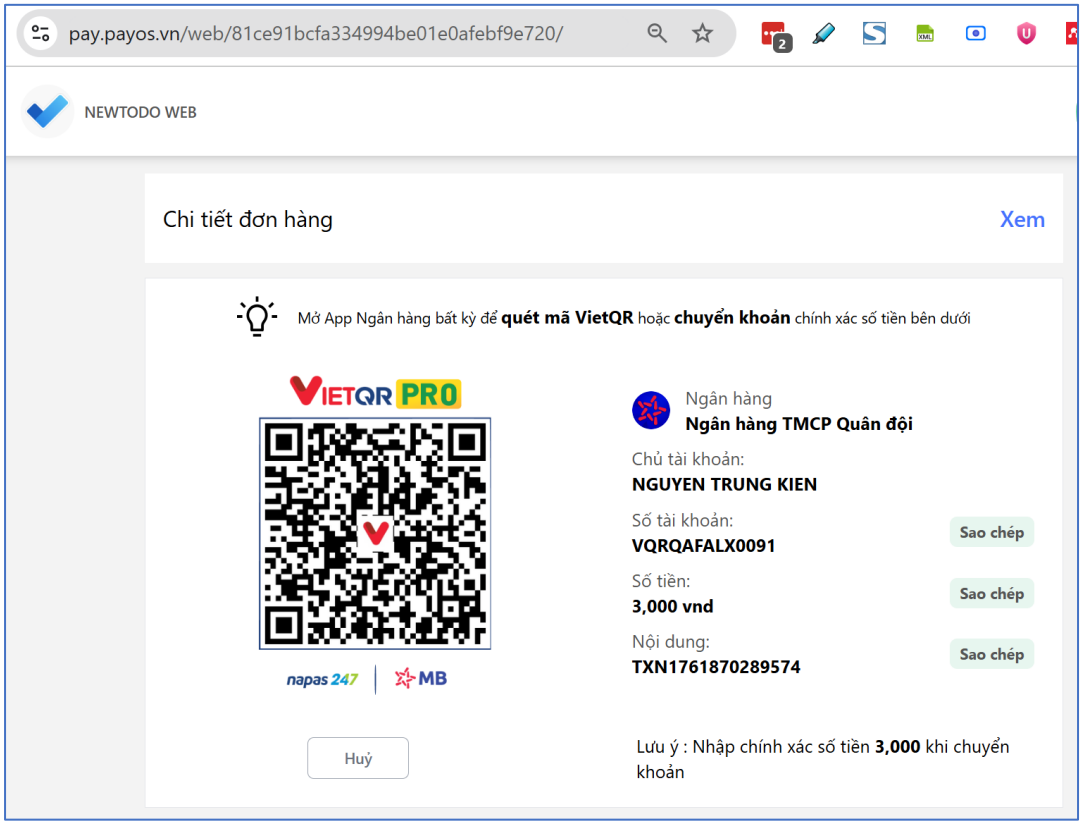
5. Course Enrollment

5.1. Learning Enroll

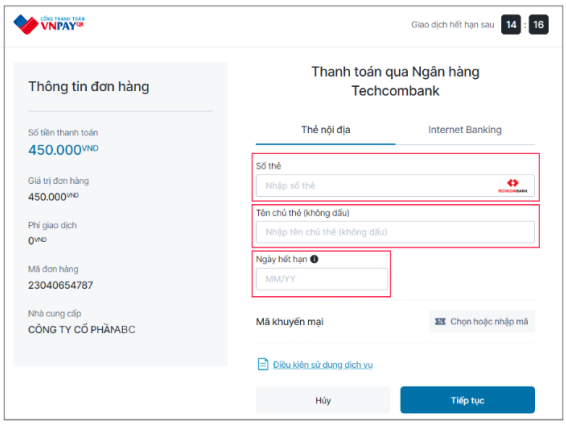


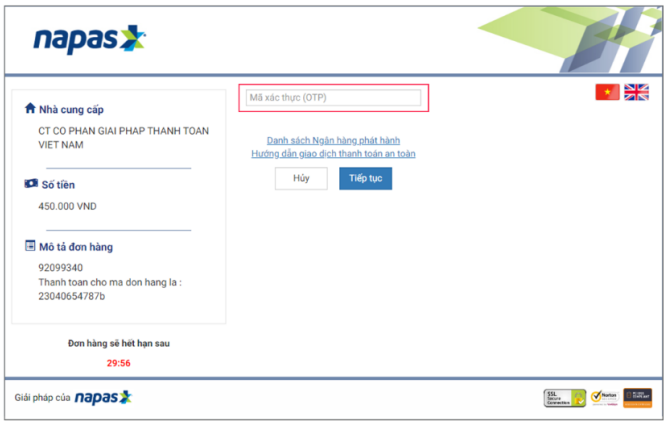
5.2. Online Payment

a. Bank Transfer (PayOS)



b. Internet Banking (VnPay)





5.3. My Enrollments

My Enrollments

Id	Course	User Info	Enrolled At	Fee($)	Status	Actions
38	AI Automation Cơ Bản với N8N – Học Nhanh, Làm Ngay	Kien Nguyen8 kien4demo@gmail.com	2025-09-30 23:55:26	68.00	Pending	Update
34	The Complete Bootcamp - error!!	Kien Nguyen8 kien4demo@gmail.com	2025-09-30 13:09:15	18.00	Approved	View
30	Khóa học thứ 4	Kien Nguyen8 kien4demo@gmail.com	2025-09-30 11:49:04	200.00	Pending	Update
28	Tiêu đề course 1	Kien Nguyen8 kien4demo@gmail.com	2025-09-30 11:43:24	0	Rejected	Update

5.4. Enrollment List

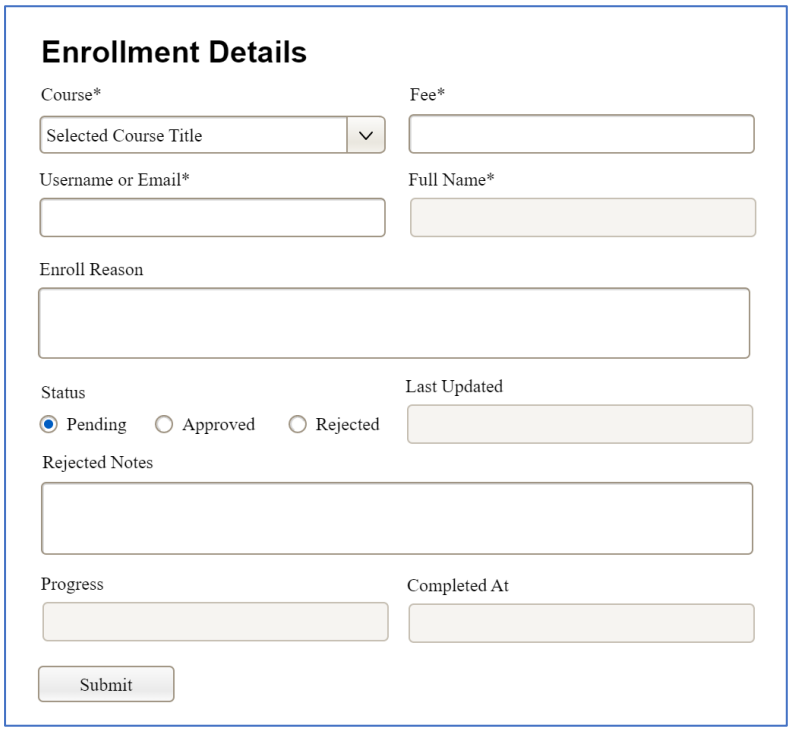
Course Enrollments

Import | Export | + Add New

All Courses | All Users | All | Search: Full Name, Email, Course | Filter

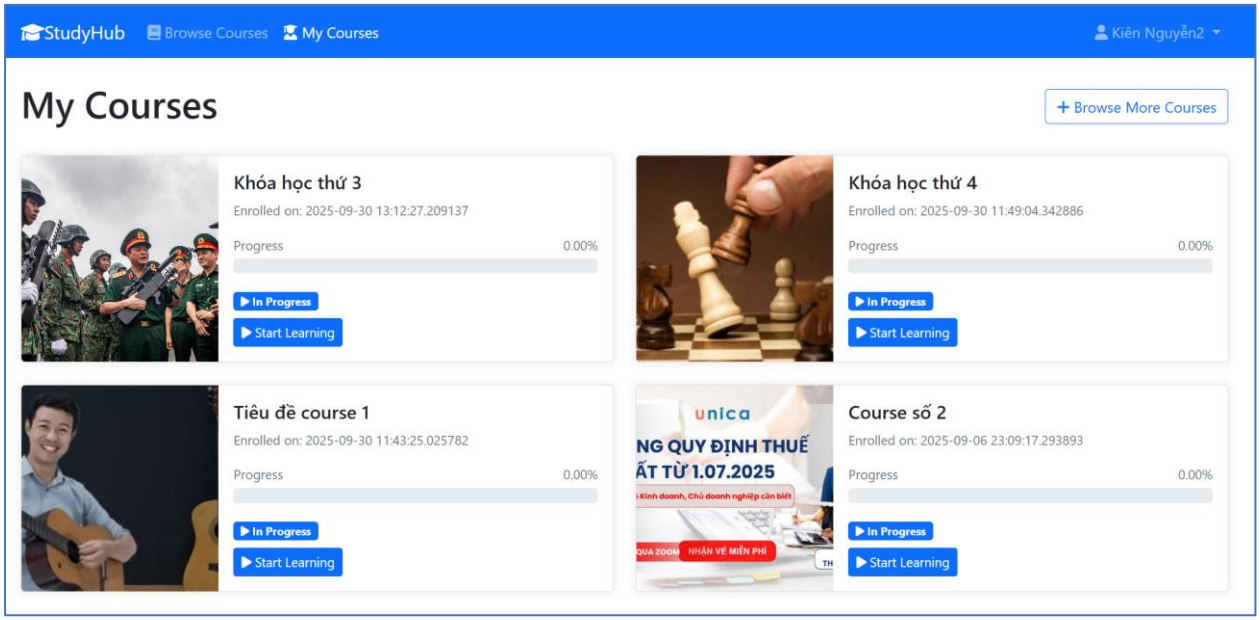
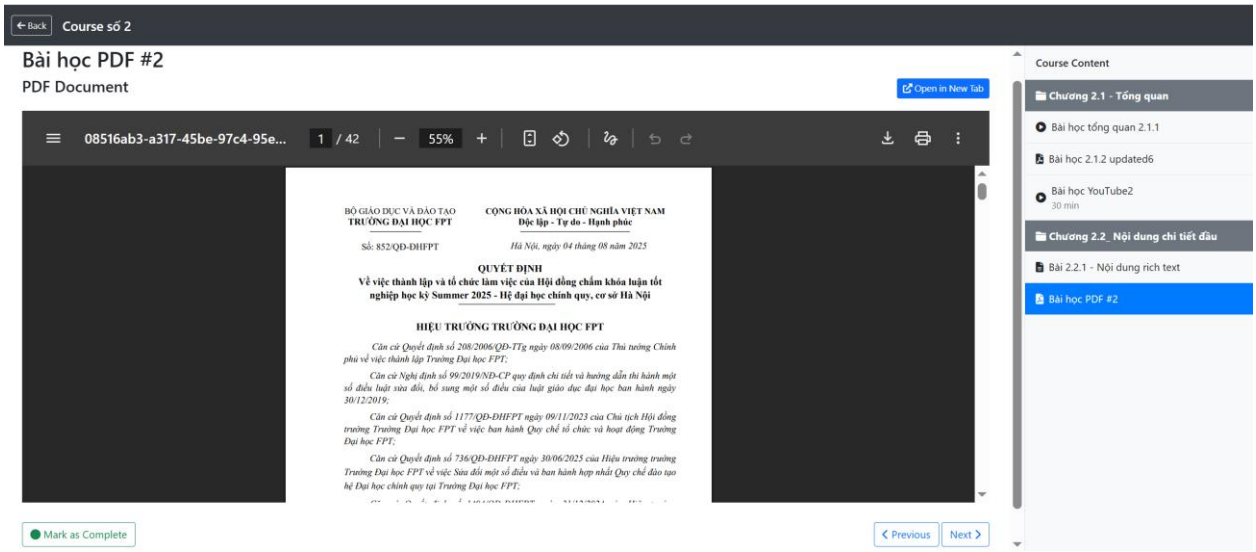
ID	Course	User Info	Enrolled At	Status	Progress	Actions
38	AI Automation Cơ Bản với N8N – Học Nhanh, Làm Ngay Fee($): 68.00	Kien Nguyen8 kien4demo@gmail.com	2025-09-30 23:55:26.92397	Pending	0.00	
37	Khóa học thứ 3 Fee($): 0.00	Kiên Nguyễn2 kienfu@gmail.com	2025-09-30 13:12:27.209137	Pending	0.00	
36	Khóa học thứ 3 Fee($): 0.00	Kien Nguyen9 kiennt@fpt.edu.vn	2025-09-30 13:12:27.113144	Pending	0.00	
34	The Complete Bootcamp - error!! Fee($): 18.00	Kien Nguyen8 kien4demo@gmail.com	2025-09-30 13:09:15.278567	Approved	0.00	
32	Khóa học thứ 4 Fee($): 200.00	Kiên Nguyễn2 kienfu@gmail.com	2025-09-30 11:49:04.342886	Approved	0.00	
30	Khóa học thứ 4 Fee($): 200.00	Kien Nguyen8 kien4demo@gmail.com	2025-09-30 11:49:04.162077	Pending	0.00	
29	Tiêu đề course 1 Fee($): 0.00	Kiên Nguyễn2 kienfu@gmail.com	2025-09-30 11:43:25.025782	Pending	0.00	
28	Tiêu đề course 1 Fee($): 0.00	Kien Nguyen8 kien4demo@gmail.com	2025-09-30 11:43:24.609816	Rejected	0.00	
27	Course số 2	Kiên Nguyễn2	2025-09-06 23:09:17.293893	Rejected	0.00	

5.5. Enrollment Details



6. Course Access

6.1. My Courses



6.2. Lessons Viewer